OpenDART

Judge quick start

Vision-driven autonomous terminal guidance with OpenCV 5.

A synthetic camera-to-control loop, with a separate perception study on recorded DART spacecraft imagery.

Try the live demonstrations

Closed-loop simulation

http://ec2-54-253-67-121.ap-southeast-2.compute.amazonaws.com/

Real DRACO imagery

http://ec2-54-253-67-121.ap-southeast-2.compute.amazonaws.com/draco

A four-step walkthrough

1. Run the nominal approach

On the simulation page, select **Nominal** and press **Run this experiment**. Allow about 15 seconds. Read the outcome and estimation error, then compare the state estimate with the clearly labelled evaluation truth.

2. Understand the controls

Review the other experiment descriptions. **Camera blackout** extends the loss of imagery. **No visual updates** and **Ballistic** are negative controls with expected misses. Applied force changes the next synthetic image.

3. Analyse a real frame

On the DRACO page, choose **About 10 s before impact** and press **Analyse this frame**. Compare the original with the new server result. Try **Final partial transmission** to see the received strip without an inferred missing body extent.

4. Read the evidence

Play **Recorded DRACO replay** and inspect **Real-image held-out benchmark**. Historical frames cannot respond to new control commands. Use the technical report for the methods, uncertainty and sample limits.

Results and limits

Headline synthetic results

96/96 guided tested cases reached contact. The campaign used two appearances, 12 seeds and four conditions. Each 12/12 condition cell has Wilson 95% interval [0.758, 1.000]. Paired conditions are not independent trials.

4/4 negative controls missed. A 40 s camera blackout was followed by an accepted update at 240 s in the tested cases.

The recorded nominal run had 9.23 m final estimation error, 557 accepted / 40 missing visual updates, 16 s navigation hold and 1.60 m/s correction. Estimation error is not residual miss distance at contact.

Headline DRACO results

18 of 20 sampled frames were manually reviewed. Autonomous tracking matched 6/8 compact points within 5 px. Median point error when selected was 1.218 px (n=6).

A visible region was available in 8/8 reviewed region cases, with median visible-region IoU 0.802 (n=8). Operator-assisted tracking and autonomous tracking have the same median IoU, 0.871, on the same six eligible later regions.

Key limitations

Synthetic contact does not establish flight readiness or statistical EKF consistency. Full-state covariance showed undercoverage. Known physical target size supplies monocular scale; the scene, attitude and actuation are simplified.

Recorded DRACO imagery supports perception replay, not closed-loop guidance validation. Two unreviewed boundary cases were excluded. There are no reviewed labels in the final ten seconds. A visible terrain region is not a physical centre or range measurement.

Read more in the extracted package

Technical methods and results: OpenDART_Technical_Report_Final.pdf

Project overview and source links: README.md

Install, run and reproduce: REPRODUCTION.md